

PREDICTIVE ANALYTICS USING HISTORICAL DATA

1. Introduction

Predictive Analytics is the process of using historical data, statistical techniques, and machine learning algorithms to predict future outcomes. This project demonstrates how historical data can be analysed to forecast future values.

2. Objective

To build a predictive model using historical data and predict future trends using Linear Regression.

3. Tools and Technologies

- Python
- Pandas
- Matplotlib
- Scikit-learn

4. Methodology

a) Data Collection

Historical data was stored in a CSV file.

b) Data Preprocessing

The dataset was cleaned by removing missing values.

c) Model Building

A Linear Regression model was trained using historical data.

d) Prediction

The trained model was used to predict future values for upcoming years.

e) Visualization

Graphs were generated to visualize historical trends and predictions.

5. Results

The model successfully predicted future values and displayed them graphically. Mean Absolute Error (MAE) was used to evaluate performance.

6. Conclusion

The project demonstrates the use of predictive analytics for forecasting future trends from historical data. Linear Regression provided a simple and effective prediction model.